



















Printer station. Firmware: reader/src/main.cpp (filament-kit).
Power: the C3's USB 5 V; every module on the C3's 3V3 pin; the buzzer on its 5V pin.
I2C: Elechouse NFC Module V3 PN532 (0x24, switch 1 ON, switch 2 OFF) and OLED (0x3C) share SDA/SCL.
PN532 IRQ and RSTO stay unconnected (firmware polls over I2C).
Encoder: A/B to GPIO 3/4 with internal pull-ups, common C to GND;
push switch S1-S2 between GPIO 1 and 3V3, internal pull-down.
Buzzer: 12085 magnetic, 42 ohm, on 5 V; BS170 Q1 low side, 220 ohm gate RG1, 100k pull-down RP1,
1N4001 flyback D1. GPIO 0 high = on, idles low.
UNCONFIRMED: PN532 supply pin.
Board: U1 plugs into two 1x8 female headers. Off-board modules plug into JST-XH headers:
J1 PN532 and J2 OLED (GND 3V3 SDA SCL), J3 encoder (A B C SW 3V3), J4 buzzer (5V BZ-).
U2, U3, SW1 and BZ1 are off the board (not placed on the PCB).

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